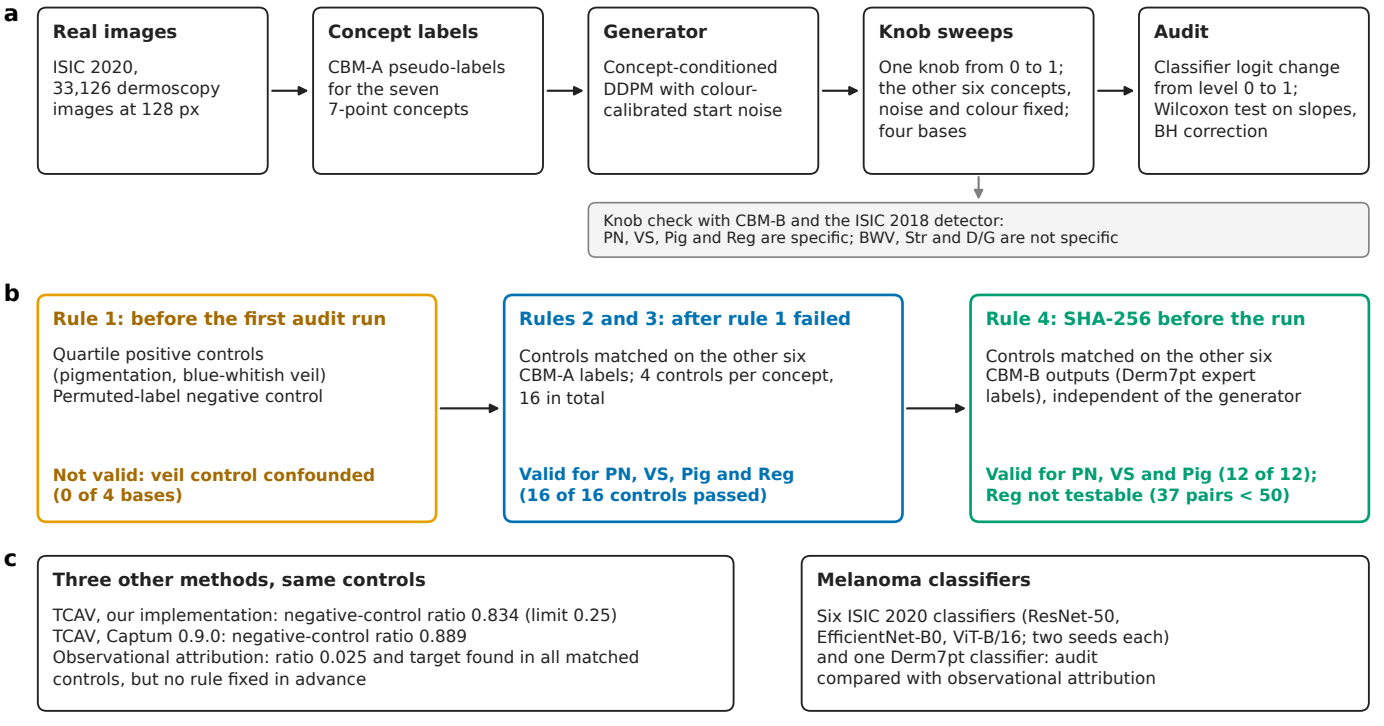
a
Real images
ISIC 2020,
33,126 dermoscopy
images at 128 px
Concept labels
CBM-A pseudo-labels
for the seven
7-point concepts
Generator
Concept-conditioned
DDPM with colour-
calibrated start noise
Knob sweeps
One knob from 0 to 1;
the other six concepts,
noise and colour fixed;
four bases
Audit
Classifier logit change
from level 0 to 1;
Wilcoxon test on slopes,
BH correction
Knob check with CBM-B and the ISIC 2018 detector:
PN, VS, Pig and Reg are specific; BWV, Str and D/G are not specific
b
Rule 1: before the first audit run
Quartile positive controls
(pigmentation, blue-whitish veil)
Permuted-label negative control
Not valid: veil control confounded
(0 of 4 bases)
Rules 2 and 3: after rule 1 failed
Controls matched on the other six
CBM-A labels; 4 controls per concept,
16 in total
Valid for PN, VS, Pig and Reg
(16 of 16 controls passed)
Rule 4: SHA-256 before the run
Controls matched on the other six
CBM-B outputs (Derm7pt expert
labels), independent of the generator
Valid for PN, VS and Pig (12 of 12);
Reg not testable (37 pairs < 50)
c
Three other methods, same controls
TCAV, our implementation: negative-control ratio 0.834 (limit 0.25)
TCAV, Captum 0.9.0: negative-control ratio 0.889
Observational attribution: ratio 0.025 and target found in all matched
controls, but no rule fixed in advance
Melanoma classifiers
Six ISIC 2020 classifiers (ResNet-50,
EfficientNet-B0, ViT-B/16; two seeds each)
and one Derm7pt classifier: audit
compared with observational attribution